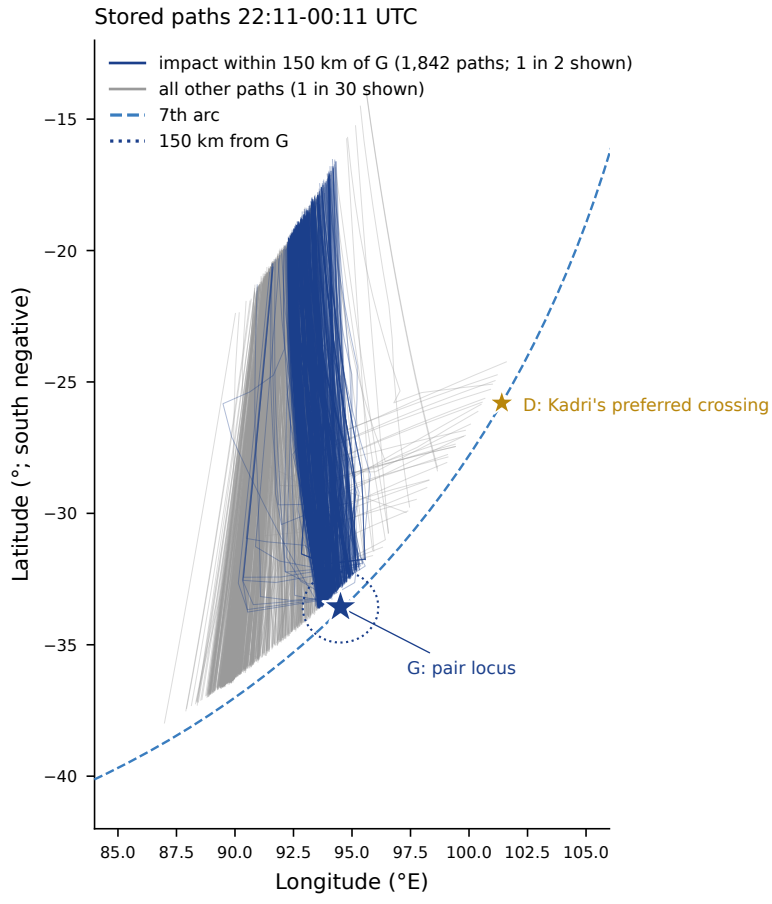
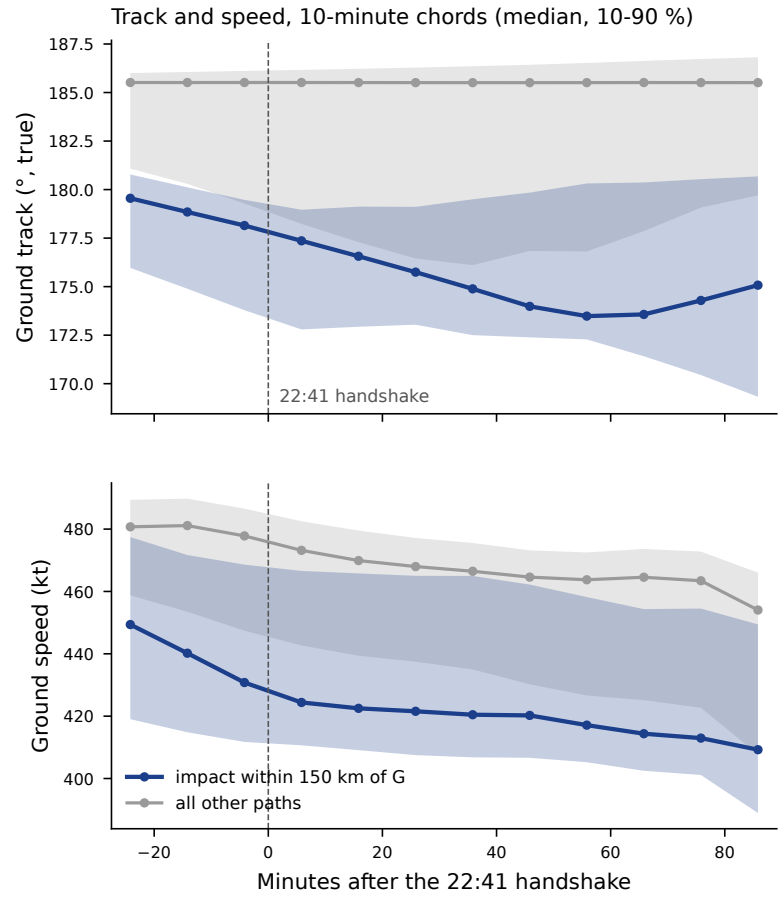
Paths that end near the pair locus fly about 10° further east of south and 30-50 kt slower after 22:41 than the other paths





(i) Left: blue lines are stored flight paths whose simulated impact is within 150 km of the pair point G; grey lines are the other paths. Right: the direction and the speed of the paths after the 22:41 satellite contact. The paths that end near G go more to the east of south and fly slower. The paths and the impacts are linked by position at 00:11, not by a direct record. The two sounds (Cape Leeuwin 00:52:04, Diego Garcia 01:12:49) have a time difference that agrees with one point on the arc, 33.6°S 94.5°E. Pairs like this occur by chance: the test gives p 0.019 from 36 tests, which is a chance level. The Diego Garcia sound is not kept when the airgun noise is removed. The sound direction that Kadri published for 00:52 does not point to this place. The impact would need about 330 times more sound than the calibration crash gives.

(ii) Pair G: Cape Leeuwin H01W 00:52:04 UTC (DIGITISED Kadri 2024 Fig. 9 panel c; envelope matched filter M1b p 0.043; Kadri's 'text time 00:52:03', 'caption time 00:54:30') x Diego Garcia South H08S 01:12:49 (DIGITISED panel e, raw, M1b p 0.039; the airgun-suppressed residual does not keep it, p_B 0.30); lag 1,245 s inside the predicted 809-1,536 s; 19 lag-consistent pairs vs 8.8 expected, permutation p 0.019 over 36 receiver-pair tests (chance level: a smallest p this low arises about half the time). Arc locus 33.6°S 94.5°E, implied impact 00:31:41; eta for P_D(both) = 0.5 is 0.069 = 330x the F-35A anchor 2.1e-4. BEARING: the locus lies at 269.0° from H01W (1,812 km) and 144.8° from H08S (3,657 km), WGS84; Kadri gives 57° (text p. 9) or 306.18° (Table 1 caption) for the 00:52 signal; neither is within 37° of the locus bearing; nearest Table 1 row 00:52:50 at 274.84° (46 s later). Paths: core run J waypoints stratum, 24 seeds x 2,000 stored route samples (routes.npy, every 600 s from 18:01:49; drawn 22:11:49-00:11:49). Each route is joined to the end-of-flight hand-off rows within 10 km of its position at 00:10:59 (POSITION association; the routes carry no row identity); f = share of those rows' D0' p 0.5 impact weight within 150 km of G. Blue: f >= 0.5 (1,842 of 48,000); bands weight each route by f (near) or 1 - f (others), sum f = 1,868. Track and speed from WGS84 chords between consecutive 600 s points (not instantaneous; a turn inside a chord is smoothed). 5th/6th-arc contours of the earlier chart are not drawn. Source posterior run J, 00:19 Held Out (D0'), dive probability p 0.5; PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, +10.4/+7.4 dB per octave at H01W/H08S) and C2 (coupling efficiency not calibrated; eta swept over four decades, F-35A upper anchor). Module branch hypothesis/hydroacoustics 9b27786; searches results-data/searches_run_j.